Lab 15: Macroevolution (Module 14)

BIOL-1

Name: ______________________ **Date:** ______________________

Overview

You are on the **Speciation Response Team**. A river canyon has split one ancestral **mock-orange fly** population into **Population North** and **Population South**. Your job is to **predict** what must happen for **one species to become two**, then diagnose **barriers** from field reports, and finally decide whether a weird plant counts as **instant speciation**. This lab is a chain of **if–then** biological puzzles—not a vocabulary list.

Learning Objectives

By the end of this lab, you should be able to:

1. Apply the biological species concept to **predict** when gene flow stops speciation cold.
2. Classify **prezygotic** vs **postzygotic** barriers and name sub-types from messy narratives.
3. Contrast **allopatric** vs **sympatric** speciation with plausible mechanisms.
4. Explain how microevolutionary change can accumulate into **reproductive isolation**.

Materials Needed

- Pencil or pen
- This worksheet

Part 1: Canyon Split — Build a Speciation Hypothesis

One species, two valleys, no bridges. Flies rarely cross the canyon.

A. Prediction: If North and South remain isolated for thousands of generations, **mutations and selection** proceed independently. **Before reading "Report B,"** answer:

Hypothesis: Will the two populations necessarily become separate species **automatically** just because they look different? **Yes / No.**
Reason:

B. Report (given): After 10,000 generations, North flies breed in **early spring**; South flies breed in **late spring**. When researchers cage them together, mating is rare; when they **force** mating, eggs are fertilized but **embryos die** before hatching.

1. List two distinct reproductive barriers visible in that report (name the barrier types—e.g., temporal, reduced hybrid viability, etc.):
1.
2.

2. Which barrier is prezygotic? Which is postzygotic?
- Prezygotic:
- Postzygotic:

C. Revision: Did your Part A hypothesis need editing? What **specific evidence** forced the change?

Part 2: Mystery Reports — Diagnose the Barrier

Each vignette is incomplete on purpose. Your diagnosis is the point.

3. For each report, choose Pre **or** Post **(prezygotic vs postzygotic).**

Field report	Pre or Post?
Pollen of Species X never starts tubes on Species Y stigmas.	
F1 hybrid salamanders survive but have **0%** successful mating as adults.	

Field report	Pre or Post?
Fireflies flash different patterns; males and females ignore the wrong pattern.	
Hybrid zygotes form, but chromosome pairing fails in early mitosis so development stops.	

4. Sub-type precision: Match report to best label (habitat / temporal / behavioral / mechanical / gametic / reduced hybrid viability / reduced hybrid fertility / hybrid breakdown).

Report (quote a few words)	Barrier sub-type
“Pollen… never starts tubes”	
“Flash different patterns”	
“F1 survive but 0% successful mating”	

Part 3: Geography — Same River, Different Endings

5. Two roads to new species:

Scenario	Allopatric or Sympatric?
Canyon splits a valley; populations diverge in isolation.	
A plant instantly inherits **4n** chromosomes and can only breed with other **4n** plants in the same meadow.	

6. Mechanism check: Why is polyploidy often **sympatric** speciation in plants? *(Hint: instant reproductive isolation from diploids.)*

7. Counterfactual: If the canyon **fills in** and flies mix freely **before** barriers evolve, what happens to speciation? **One paragraph (4–5 sentences):**

Part 4: From Lab 14 to Today — Micro Builds Macro

No new "evolution force" appears at the speciation threshold—only barriers to gene flow.

8. Chain argument: Starting from **mutation** creating variation in each isolated population, explain how **natural selection** and **drift** could make **prezygotic** barriers more likely over time. (4–5 sentences)

9. Final verdict: Under the biological species concept, if two populations still have **fertile hybrids whenever they meet**, are they definitely two species? **Explain the nuance** (gene flow vs lineage divergence) in 3–4 sentences.